

Algebra II with Trigonometry

Chapter 7.4 & 7.5

Olympiad Problems (Gemini)

Saved tutoring response with MathJax rendering. Original PDF file:
[4626701_alg-2trig-h-chp-7-4-7-5-note-docx.pdf](#)

Problems

1. Find the sum of all distinct solutions $x \in [0, 2\pi)$ to the equation $\sin x + \sin 2x + \sin 3x = 0$.
2. Evaluate the exact numerical value of the following expression without a calculator: $\tan 20^\circ + \tan 40^\circ + \sqrt{3} \tan 20^\circ \tan 40^\circ$
3. Given that $\tan x + \cot x = 4$, determine the exact value of $\sin^4 x + \cos^4 x$.
4. Find all real solutions $x \in [0, 2\pi)$ that satisfy the equation $\sin^8 x + \cos^8 x = 1$.
5. Evaluate the exact value of $\sec^4\left(\frac{\pi}{12}\right) + \csc^4\left(\frac{\pi}{12}\right)$.
6. Determine the unique angle $x \in (0, \frac{\pi}{2})$ that satisfies the equation $\cos x + \sin x = \sqrt{2} \cos\left(\frac{x}{2}\right)$.